



DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

CLASS

INDEX
NUMBER

PRELIMINARY EXAMINATION 2024 SECONDARY 4 EXPRESS / 5 NORMAL ACADEMIC

ADDITIONAL MATHEMATICS

Paper 2

4049/02

22 August 2024
2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Calculator Model

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved scientific calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 90.

Total Marks

Mathematical Formulae**1. ALGEBRA***Quadratic Equation*

For the equation $ax^2 + bx + c = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{r}a^{n-r}b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\dots(n-r+1)}{r!}$.

2. TRIGONOMETRY*Identities*

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for ΔABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Area of } \Delta = \frac{1}{2} ab \sin C$$

Answer **all** the questions.

- 1 Solve the equation $\log_p x + 2\log_x p = 3$, where p is a positive constant, giving x in terms of p . [5]

Marking Scheme

$$\log_p x + \frac{2}{\log_p x} = 3 \quad \text{--- B1 (For an appropriate change of base)}$$

$$(\log_p x)^2 + 2 = 3\log_p x$$

$$(\log_p x)^2 - 3\log_p x + 2 = 0$$

--- M1 (For an attempt to obtain a 3-term quadratic equation in terms of $\log_p x$ or $\log_x p$, equate to zero)

$$(\log_p x - 1)(\log_p x - 2) = 0$$

$$\log_p x - 1 = 0 \text{ or } \log_p x - 2 = 0 \quad \text{--- M1 (Correct solution if *their* quadratic equation)}$$

$$\log_p x = 1 \text{ or } \log_p x = 2$$

$$x = p \text{ or } x = p^2 \quad \text{--- A2 (A1 each, must be from completely correct work)}$$

3 (a) A curve has equation $y = \frac{e^{2x}}{x+2}$, where $x > -2$.

(i) Find $\frac{dy}{dx}$. [2]

Marking Scheme

$$\begin{aligned} & \frac{dy}{dx} \\ &= \frac{2e^{2x}(x+2) - 1(e^{2x})}{(x+2)^2} \quad \text{--- B2 (B1 for numerator, B1 for denominator)} \\ &= \frac{e^{2x}(2x+3)}{(x+2)^2} \end{aligned}$$

(ii) Show that the y-coordinate of the turning point is $\frac{2}{e^3}$. [2]

Marking Scheme

$$\begin{aligned} & \frac{e^{2x}(2x+3)}{(x+2)^2} = 0 \\ & e^{2x} = 0 \text{ (rejected),} \\ & 2x+3 = 0 \\ & x = -1\frac{1}{2} \quad \text{--- M1 (Reject } e^{2x} = 0 \text{ and find } x, \text{ equate their } \frac{dy}{dx} = 0) \end{aligned}$$

$$y = \frac{e^{2(-1.5)}}{-1.5+2} = \frac{2}{e^3} \quad \text{--- A1 (cao)}$$

(iii) Determine the nature of the turning point. [2]

Marking Scheme

x	-1.6	-1.5	-1.4
$\frac{dy}{dx}$	-ve	0	+ve
gradient			

--- B1 First derivative test

It is a minimum point. --- B1 (cao, conclusion based on first derivative test)

3 (a) Solve the equation $4^x - 4(2^{x+1}) + 7 = 0$.

[4]

Marking Scheme

$$2^{2x} - 4(2^x \bullet 2) + 7 = 0 \quad \text{--- M1 (either } 4^x = 2^{2x} \text{ or } 2^{x+1} = 2^x \bullet 2)$$

$$2^{2x} - 8(2^x) + 7 = 0 \quad \text{--- M1 (Quadratic equation)}$$

$$(2^x - 7)(2^x - 1) = 0$$

$$2^x = 7 \quad \text{or} \quad 2^x = 1$$

$$x = \frac{\lg 7}{\lg 2} = 2.81 \quad \text{or} \quad x = 0 \quad \text{--- A2 (A1 each)}$$

- (b) As people get vaccinated, the number of active cases of people infected with a virus reduces by half each day. Vaccination started at the beginning of 5 February 2021. Given that the number of active cases at the beginning of 5 February 2021 was 59649, find the number of active cases, to the nearest whole number, at the beginning of 12 February 2021. [2]

Marking Scheme

5 Feb is Day 0, 12 Feb is Day 7

Let number of active cases be A and no. of days be d .

$$A = 59649 \left(\frac{1}{2} \right)^d \quad \text{--- M1 (knowing that } \left(\frac{1}{2} \right)^d, \text{ or doing manually reduced by multiplying by } \frac{1}{2} \text{ 7 times)}$$

$$\text{No. of active cases on 12 Feb} = 59649 \left(\frac{1}{2} \right)^7 = 466 \quad \text{--- A1}$$

- (c) A liquid is allowed to cool after being heated. The temperature, $\theta^\circ\text{C}$, of the liquid, t seconds after being removed from the heat, is given by $\theta = 20 + 80e^{-0.01t}$. Collin claimed that the temperature of the liquid after a long time will approach the temperature 20°C . Explain why he is correct. [2]

Marking Scheme

$$t \rightarrow \infty, e^{-0.01t} \rightarrow 0 \quad \text{--- B1}$$

$$\text{Therefore } 20 + 80e^{-0.01t} \rightarrow 20$$

Hence Collin is correct.

- 4 (a)** Show that the equation $x^2 + (2 - k)x + k = 3$ has real roots for all real values of k . [3]

Marking Scheme

Discriminant

$$\begin{aligned}
 &= (2 - k)^2 - 4(1)(k - 3) \quad \text{--- B1 (Form discriminant)} \\
 &= 4 - 4k + k^2 - 4k + 12 \\
 &= k^2 - 8k + 16 \\
 &= (k - 4)^2 \quad \text{--- B1 (Show perfect square)}
 \end{aligned}$$

Since $(k - 4)^2 \geq 0$, therefore discriminant ≥ 0 , hence equation has real roots for all real values of k . --- B1

- (b)** Find the smallest value of the integer a for which $ax^2 + 5x + 2$ is positive for all values of x . [3]

Discriminant < 0

$$\begin{aligned}
 5^2 - 4(a)(2) &< 0 \quad \text{--- M2 (M1 for Discriminant correctly sub in and M1 for Discriminant } < 0)} \\
 25 - 8a &< 0 \\
 a &> 3\frac{1}{8}
 \end{aligned}$$

Smallest integer $a = 4$ --- A1

- (c)** Find the values of p and q for which $\{x: x < -3 \text{ or } x > 5\}$ is the solution set of $x^2 - px > q$. [2]

Marking Scheme

$$\begin{aligned}
 x &< -3 \text{ or } x > 5 \\
 (x + 3)(x - 5) &> 0 \quad \text{--- M1 (Form quadratic inequality correctly)} \\
 x^2 - 2x &> 15 \\
 p = 2 \text{ and } q = 15 &\quad \text{--- A1 (both answers are correct)}
 \end{aligned}$$

5 The equation of a polynomial is given by $p(x) = x^3 - x^2 + ax + 9$.

(a) Given that $x - 1$ is a factor of $p(x)$, show that $a = -9$.

[2]

Marking Scheme

$$1^3 - 1^2 + a(1) + 9 = 0 \quad \text{--- B2 (B1: Sub } x = 1 \text{ into polynomial, B1: Equate to 0)}$$

$$a = -9 \text{ (If this is not seen, minus 1 mark)}$$

(b) Solve $x^3 - x^2 - 9x + 9 = 0$.

[3]

Marking Scheme

$$x^3 - x^2 - 9x + 9 = (x - 1)(x^2 + px + q)$$

$$\text{Comparing coefficients of } x^2, p - 1 = -1, \text{ therefore } p = 0$$

$$\text{Comparing constant terms, } q = -9$$

$$x^3 - x^2 - 9x + 9 = (x - 1)(x^2 - 9) \quad \text{--- M1 (Obtain quadratic factor by inspection method or long division)}$$

$$x^3 - x^2 - 9x + 9 = 0$$

$$(x - 1)(x + 3)(x - 3) = 0 \quad \text{--- M1 (method to find } x)$$

$$x = 1, -3 \text{ or } 3 \quad \text{--- A1 (all 3 answers)}$$

(c) Hence solve the equation $\cot^3(2y + 30^\circ) - \cot^2(2y + 30^\circ) - 9\cot(2y + 30^\circ) + 9 = 0$
for $0^\circ < y < 180^\circ$.

[4]

Marking Scheme

$$\cot(2y + 30^\circ) = 1, 3, -3$$

$$\tan(2y + 30^\circ) = 1, \frac{1}{3}, -\frac{1}{3}$$

--- M1 for tan ratio equated to the various values

$$\tan(2y + 30^\circ) = 1$$

$$2y + 30^\circ = 45^\circ, 225^\circ$$

$$y = 7.5^\circ, 97.5^\circ \text{ --- A1 (both answers)}$$

$$\tan(2y + 30^\circ) = \frac{1}{3}, -\frac{1}{3}$$

Angle in all 4 quadrants

$$\tan \alpha = \frac{1}{3}$$

$$\alpha = 18.434^\circ \text{ --- M1}$$

$$2y + 30^\circ = 18.434^\circ, 161.566^\circ, 198.434^\circ, 341.566^\circ, 378.434^\circ$$

$$y = 65.8^\circ, 84.2^\circ, 155.8^\circ, 174.2^\circ \text{ --- A1 (all 4 answers)}$$

6 (a) Show that $\frac{d}{dx} \left[x(2x+3)^{\frac{3}{2}} \right] = (5x+3)\sqrt{2x+3}$. [5]

Marking Scheme

$$\begin{aligned} & \frac{d}{dx} \left[x(2x+3)^{\frac{3}{2}} \right] \\ &= (2x+3)^{\frac{3}{2}} + \frac{3}{2}(2x+3)^{\frac{1}{2}} \bullet 2 \bullet x \quad \text{--- B2} \\ &= (2x+3)^{\frac{3}{2}} + 3x(2x+3)^{\frac{1}{2}} \\ &= (2x+3)^{\frac{1}{2}} (2x+3+3x) \quad \text{--- M2} \\ &= (5x+3)\sqrt{2x+3} \quad \text{---A1} \end{aligned}$$

(b) Hence find $\int x\sqrt{2x+3} \, dx$, giving your answer in the form $k(x-1)(2x+3)^{\frac{3}{2}} + c$, where k is a constant to be found and c is a constant of integration which cannot be found. [4]

Marking Scheme

$$\begin{aligned} x(2x+3)^{\frac{3}{2}} &= \int (5x+3)\sqrt{2x+3} \, dx \quad \text{--- M1} \\ x(2x+3)^{\frac{3}{2}} &= \int 5x\sqrt{2x+3} + 3\sqrt{2x+3} \, dx \quad \text{--- M1} \\ x(2x+3)^{\frac{3}{2}} &= 5 \int x\sqrt{2x+3} \, dx + 3 \int (2x+3)^{\frac{1}{2}} \, dx \\ x(2x+3)^{\frac{3}{2}} &= 5 \int x\sqrt{2x+3} \, dx + 3 \bullet \frac{(2x+3)^{\frac{3}{2}}}{2\left(\frac{3}{2}\right)} + c \quad \text{--- M1 (correct integration)} \\ 5 \int x\sqrt{2x+3} \, dx &= x(2x+3)^{\frac{3}{2}} - (2x+3)^{\frac{3}{2}} + c \\ \int x\sqrt{2x+3} \, dx &= \frac{1}{5} \left[x(2x+3)^{\frac{3}{2}} - (2x+3)^{\frac{3}{2}} \right] + C \\ &= \frac{1}{5} (x-1)(2x+3)^{\frac{3}{2}} + C \quad \text{---A1} \end{aligned}$$

- 7 A circle, with centre C , passes through the points $P(2, -2)$ and Q .
The point $M(0, 2)$ is the midpoint of the chord PQ and the line CQ is parallel to the x -axis.

(a) Show that C is $(8, 6)$.

[5]

Marking Scheme

Let Q be (a, b)

Midpoint of $PQ = M$

$$\left(\frac{2+a}{2}, \frac{-2+b}{2} \right) = (0, 2)$$

Therefore, $a = -2, b = 6$

$Q(-2, 6)$ --- M1

y -coordinate of $C = 6$. (Given that CQ is parallel to x -axis)

$$\text{Gradient of } PM = \frac{-2-2}{2-0} = -2 \quad \text{--- M1}$$

$$\text{Gradient of } CM = \frac{1}{2} \quad \text{--- M1}$$

$$y = \frac{1}{2}x + c$$

$$\text{Sub } M(0, 2): 2 = \frac{1}{2}(0) + c$$

$$c = 2$$

$$\text{Equation of line } CM: y = \frac{1}{2}x + 2 \quad \text{--- M1}$$

Sub $y = 6$:

$$6 = \frac{1}{2}x + 2$$

$$x = 8$$

$C(8, 6)$ --- A1

(b) Find the equation of the circle.

[2]

Marking Scheme

$$\text{Radius } CP = \sqrt{(8-2)^2 + (6-(-2))^2} = 10 \quad \text{--- M1}$$

Equation of circle:

$$(x-8)^2 + (y-6)^2 = 100 \quad \text{--- A1}$$

Accept this answer too: $x^2 + y^2 - 16x - 12y = 0$

(c) PR is a diameter of the circle and S is the point $(-1, 13)$.
Determine if the angle PSR is acute or obtuse.

[3]

Marking Scheme

Radius $CP = 10$ units

$$CS = \sqrt{(8-(-1))^2 + (6-13)^2} = 11.401 \quad \text{--- M1}$$

Since $CS > \text{radius } 10$ units, S is outside the circle. --- M1

Hence angle PSR is acute. --- A1

- 8 A particle moves in a straight line so that, t seconds after passing through O , its velocity, v cm/s, is given by $v = t^2 - 7t + 6$. The particle comes to instantaneous rest, firstly at A and then at B .

- (a) Find an expression, in terms of t , for the distance of the particle, s , from O at time t . [1]

Marking Scheme

$$\begin{aligned} s &= \int v \, dt \\ &= \frac{t^3}{3} - \frac{7t^2}{2} + 6t + c \\ c &= 0 \\ s &= \frac{t^3}{3} - \frac{7t^2}{2} + 6t \quad \text{--- B1} \end{aligned}$$

- (b) Calculate the distance AB . [4]

Marking Scheme

Instantaneous rest, $v = 0$.

$$\begin{aligned} t^2 - 7t + 6 &= 0 \quad \text{--- M1} \\ (t-6)(t-1) &= 0 \\ t &= 1 \text{ or } 6 \quad \text{--- M1 (method of solving seen)} \end{aligned}$$

$$\text{At A, } s = \frac{1^3}{3} - \frac{7(1)^2}{2} + 6(1) = 2\frac{5}{6}$$

$$\text{At B, } s = \frac{6^3}{3} - \frac{7(6)^2}{2} + 6(6) = -18$$

Either A or B displace correct, M1

$$\text{Distance AB} = 20\frac{5}{6} \text{ units. --- A1}$$

- (c) Find the total distance moved by the particle in the first 6 seconds after passing through O . [2]

Marking Scheme

$$\begin{aligned} \text{Total distance} &= 20\frac{5}{6} + 2\frac{5}{6} \quad \text{--- M1} \\ &= 23\frac{2}{3} \text{ units --- A1} \end{aligned}$$

- (d) Given that C is the point at which the particle has zero acceleration, determine, with full working, whether C is nearer to O or to B . [4]

Marking Scheme

$$a = \frac{dv}{dt} = 2t - 7 \quad \text{--- M1}$$

$$a = 0$$

$$2t - 7 = 0$$

$$t = 3.5 \quad \text{--- M1}$$

$$s = \frac{3.5^3}{3} - \frac{7(3.5)^2}{2} + 6(3.5) = -7\frac{7}{12} \quad \text{--- M1}$$

$$OC = 7\frac{7}{12} \text{ units}$$

$$BC = 18 - 7\frac{7}{12} = 10\frac{5}{12} \text{ units}$$

C is nearer to O . --- A1 (with OC and BC lengths seen)

- 9 (a) By considering the general term in the binomial expansion of $\left(x^2 - \frac{2}{x}\right)^7$, explain why every term is dependent on x . [3]

Marking Scheme

General term

$$= {}^7C_r (x^2)^{7-r} \left(-\frac{2}{x}\right)^r \quad \text{--- M1}$$

$$= {}^7C_r \bullet (-2)^r \bullet x^{14-2r-r}$$

Term not independent of x , power of $x = 0$.

$$14 - 3r = 0$$

$$r = 4\frac{2}{3} \quad \text{--- M1}$$

Since r is not an integer, therefore every term is independent of x . --- A1

- (b) The first three terms, in ascending powers of x , in the expansion of $\left(1 + \frac{x}{6}\right)^n (2 - 3x)^3$ can be written in the form $8 - 20x + px^2$, where n and p are constants and $n > 0$. Find the values of n and p . [9]

Marking Scheme

$$\left(1 + \frac{x}{6}\right)^n (2 - 3x)^3$$

$$= \left(1 + n\left(\frac{x}{6}\right) + \frac{n(n-1)}{1 \times 2} \left(\frac{x}{6}\right)^2 + \dots\right) (8 - 36x + 54x^2)$$

--- M5 (M1 for each term in first bracket, M1 for every one to two correct terms in second bracket)

$$\text{Coefficients of } x \text{ term} = \frac{8n}{6} - 36 \quad \text{--- M1}$$

$$\frac{8n}{6} - 36 = -20 \quad \text{--- A1}$$

$$n = 12$$

$$54 - \frac{36(12)}{6} + \frac{8(12)(11)}{72} = p \quad \text{--- M1}$$

$$p = -3\frac{1}{3} \quad \text{--- A1}$$

- 10 (a) It is known that the variables x and y are related by the equation $y = \frac{p}{x - q}$, where p and q are unknown constants.
- (i) Express this equation in a form suitable for drawing a straight-line graph, and state which variable should be used for each axis. [2]

Marking Scheme

Accept any correct answers, here are 2 solutions.

Solution 1:

$$y(x - q) = p$$

$$yx - yq = p$$

$$xy = yq + p \quad \text{--- M1}$$

y is the horizontal axis, and xy is the vertical axis. ---A1

Solution 2:

$$x - q = \frac{p}{y}$$

$$x = \frac{p}{y} + q \quad \text{--- M1}$$

$\frac{1}{y}$ is the horizontal axis, and x is the vertical axis. ---A1

- (ii) Explain how the value of p and of q could be determined from this graph. [2]

Marking Scheme

Based on Solution 1: q = gradient and p = vertical intercept

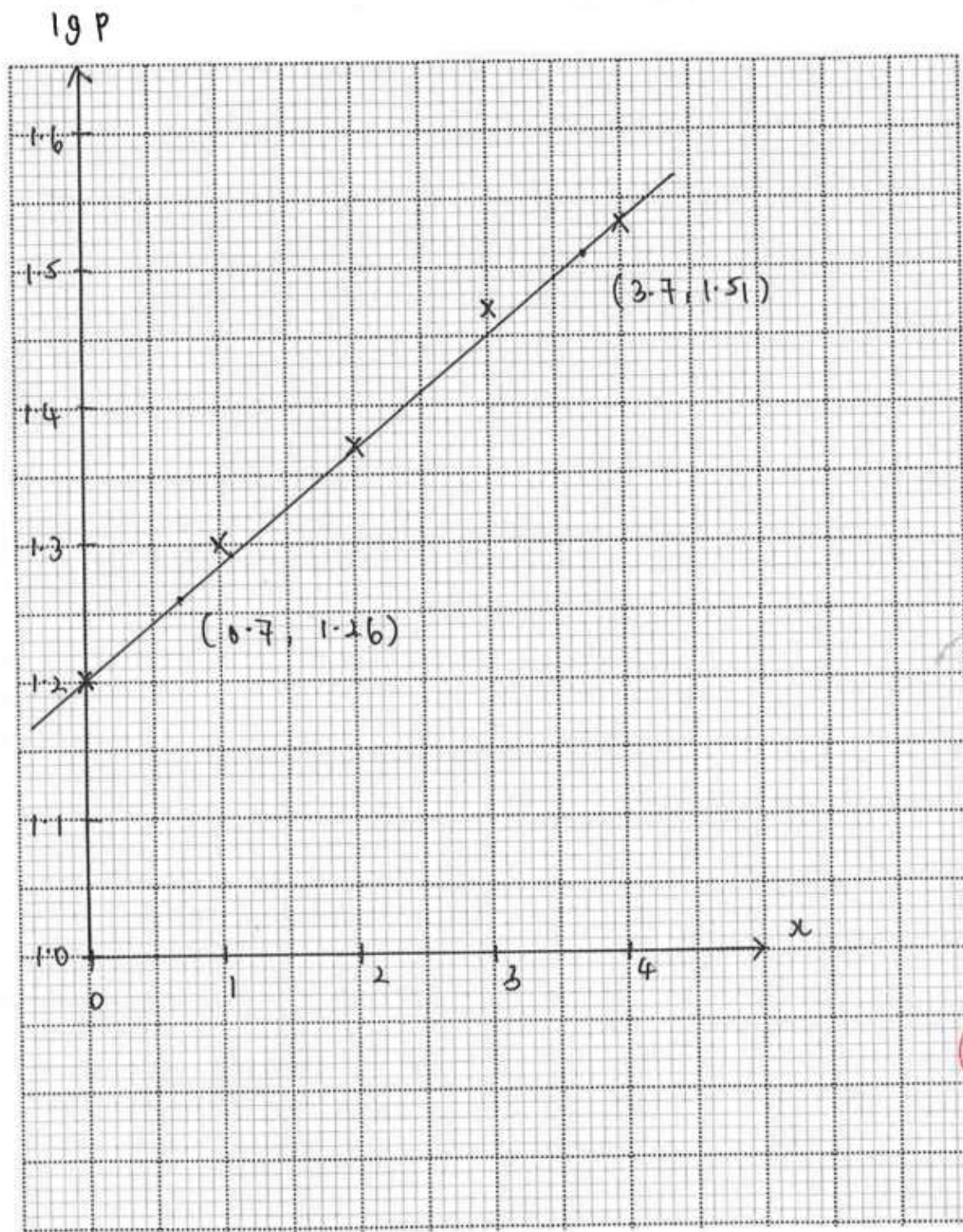
Based on Solution 2: p = gradient and q = vertical intercept.

- (b) The table shows, to 2 decimal places, the population, P , in millions, of California on January 1st at intervals of 10 years, from 1960 to 2000. The variable x is measured in units of 10 years.

Year	1960	1970	1980	1990	2000
x	0	1	2	3	4
P	15.72	19.97	23.67	29.76	33.87

- (i) On the grid on page ?? plot $\lg P$ against x and draw a straight line graph. [2]

$\lg p$ 1.20 1.30 1.37 1.47 1.53



All points plotted correctly --- P1

Straight line (best fit) --- S1

- (iii) Find the gradient of your straight line and hence express P in the form Ab^x , where A and b are constants. [4]

Marking Scheme

Use points on the line (but not given in table) to find gradient:

Gradient

$$= \frac{1.51 - 1.26}{3.7 - 0.7}$$

$$= \frac{1}{12} \text{ or } 0.0833 \text{ --- B1}$$

Vertical intercept = 1.2 (from graph)

$$\lg p = 0.0833x + 1.20$$

$$P = Ab^x$$

$$\lg P = (\lg b)x + \lg A \text{ ---B1}$$

$$\lg b = 0.0833$$

$$b = 1.21$$

$$\lg A = 1.2$$

$$A = 15.8$$

$$P = 15.8(1.21)^x \text{ --- A2}$$

- (iv) The population of California in 2020 was estimated to be 39.37 million. Explain, with mathematical workings, whether if the model found in (iii) remains valid. [2]

In year 2020, $x = 6$.

$$P = 15.8(1.21)^6 = 49.59 \text{ mil --- M1}$$

The model is no longer valid because 49.49 mil is much bigger than 39.37 mil. --- A1